⚙️ Setup & Initialization

1️⃣ Hardware Setup

- Connect the **LPC2148 ARM7** microcontroller.
- Connect the **16×2 LCD** for system information and status display.
- Connect the **4×4 Keypad** for user configuration.
- Connect the **RTC** for time and date management.
- Connect **LEDs and Buzzer** for hydration reminders.
- Connect the **Drink/Acknowledge Push Button** for recording water consumption.

2️⃣ Software Setup

- Install **Keil µVision** with ARM7/LPC2148 support.
- Create/open the project in Keil.
- Add all required source and header files:
- `main.c`
- `lcd.c / lcd.h`
- `kpm.c / kpm.h`
- `rtc.c / rtc.h`
- `hydration.c / hydration.h`
- `reminder.c / reminder.h`
- `display.c / display.h`
- Required delay, buzzer, and interrupt files.

3️⃣ Project Configuration

- Select the **LPC2148** target device.
- Configure the required **CPU clock frequency**.
- Configure GPIO pins for LCD, keypad, LEDs, buzzer, and push button.
- Initialize the **RTC** with the required time and date.
- Initialize the **LCD, keypad, interrupts, LEDs, and buzzer**.

4️⃣ Build & Program

1. Build the project in Keil.
2. Resolve any compilation or linker errors.
3. Generate the `.hex` file.
4. Load the HEX file into the LPC2148 using the required programmer/simulator.

5️⃣ System Startup

After reset, the system performs initialization and displays the welcome screen:

```
┌──────────────────┐
│   AQUAGUARDIAN   │
```

```
│       SYSTEM ON        │
└────────────────────────┘
```

The system then enters the **main monitoring screen**, where the RTC time, hydration goal, consumed glasses, remaining glasses, and hydration percentage are displayed.